week03_NLP_EDA

February 6, 2026

1 Setup

[21]:
```
# 1) Go to repo
%cd /home/jupyter-guotq/HelpHerInvest

# 2) Sync latest from GitHub before editing
!git pull --rebase origin main
```

```
/home/jupyter-guotq/HelpHerInvest
From https://github.com/tongyuguo/HelpHerInvest
 * branch            main       -> FETCH_HEAD
Already up to date.
```

[22]:
```
import os
import zipfile
from pathlib import Path
from datetime import datetime

import re

import numpy as np
import pandas as pd

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.feature_extraction.text import CountVectorizer, TfidfVectorizer
from sklearn.neighbors import NearestNeighbors

pd.set_option("display.max_columns", 200)
pd.set_option("display.max_colwidth", 180)
sns.set_theme(style="whitegrid")

repo_dir = "."

# list available data files
!ls -lah {repo_dir}/Data
```

```python
# Create Data directory path
DATA_DIR = Path(repo_dir) / "Data"

# Load the dataset
dataset_path= DATA_DIR / "stock_symbols_new.csv.zip"
print(dataset_path)
csv_file_name = "stock_symbols_new.csv"


with zipfile.ZipFile(dataset_path, 'r') as zf:
    with zf.open(csv_file_name) as file_handle:
        # Read the file-like object directly into pandas
        df = pd.read_csv(file_handle)

print("Shape:", df.shape)
df.head(3)
```

```
total 17M
drwxr-xr-x 2 jupyter-guotq jupyter-guotq  186 Feb  6 01:58 .
drwxr-xr-x 9 jupyter-guotq jupyter-guotq 4.0K Feb  6 03:00 ..
-rw-r--r-- 1 jupyter-guotq jupyter-guotq 2.4M Jan 31 01:15
dependent_variables.csv
-rw-r--r-- 1 jupyter-guotq jupyter-guotq 2.8M Feb  6 01:58 final_dataset.csv
-rw-r--r-- 1 jupyter-guotq jupyter-guotq 1.1M Feb  6 01:58 final_dataset.csv.zip
-rw-r--r-- 1 jupyter-guotq jupyter-guotq 674K Jan 31 01:15
independent_variables.csv
-rw-r--r-- 1 jupyter-guotq jupyter-guotq 9.2M Jan 29 18:07
stock_symbols_new.csv.zip
-rw-r--r-- 1 jupyter-guotq jupyter-guotq 296K Jan 30 21:22 testing_small.csv.zip
Data/stock_symbols_new.csv.zip
Shape: (10284, 224)
```

```
/tmp/ipykernel_1864996/788425580.py:38: DtypeWarning: Columns
(54,204,212,213,214,220,221,222,223) have mixed types. Specify dtype option on
import or set low_memory=False.
  df = pd.read_csv(file_handle)
```

[22]:

```
  symbol   company_name                   address1           city state  \
0   NVDA    NVIDIA CORP  2788 San Tomas Expressway    Santa Clara    CA
1  GOOGL  Alphabet Inc.  1600 Amphitheatre Parkway  Mountain View    CA
2   AAPL     Apple Inc.         One Apple Park Way      Cupertino    CA

     zip        country           phone                website  \
0  95051  United States    408 486 2000  https://www.nvidia.com
1  94043  United States    650-253-0000         https://abc.xyz
2  95014  United States  (408) 996-1010   https://www.apple.com

                           industry                    industryKey  \
```

```
0                  Semiconductors                semiconductors
1  Internet Content & Information  internet-content-information
2            Consumer Electronics          consumer-electronics

                     industryDisp                   sector  \
0                  Semiconductors               Technology
1  Internet Content & Information  Communication Services
2            Consumer Electronics               Technology

                sectorKey              sectorDisp  \
0              technology              Technology
1  communication-services  Communication Services
2              technology              Technology

   longBusinessSummary  \
0  NVIDIA Corporation, a computing infrastructure company, provides graphics and
compute and networking solutions in the United States, Singapore, Taiwan, China,
Hong Kong, and in…
1  Alphabet Inc. offers various products and platforms in the United States,
Europe, the Middle East, Africa, the Asia-Pacific, Canada, and Latin America. It
operates through Goog…
2  Apple Inc. designs, manufactures, and markets smartphones, personal
computers, tablets, wearables, and accessories worldwide. The company offers
iPhone, a line of smartphones; …

   fullTimeEmployees  \
0            36000.0
1           190167.0
2           166000.0

       companyOfficers  \
0  [{'maxAge': 1, 'name': 'Mr. Jen-Hsun  Huang', 'age': 62, 'title': 'Co-
Founder, CEO, President & Director', 'yearBorn': 1963, 'fiscalYear': 2025,
'totalPay': 11054945, 'exercise…
1  [{'maxAge': 1, 'name': 'Mr. Sundar  Pichai', 'age': 52, 'title': 'CEO &
Director', 'yearBorn': 1973, 'fiscalYear': 2024, 'totalPay': 10319413,
'exercisedValue': 0, 'unexercised…
2  [{'maxAge': 1, 'name': 'Mr. Timothy D. Cook', 'age': 64, 'title': 'CEO &
Director', 'yearBorn': 1961, 'fiscalYear': 2025, 'totalPay': 16759518,
'exercisedValue': 0, 'unexercise…

   auditRisk  boardRisk  compensationRisk  shareHolderRightsRisk  overallRisk  \
0        5.0       10.0               4.0                    8.0          8.0
1        7.0        9.0              10.0                   10.0         10.0
2        7.0        1.0               3.0                    1.0          1.0

   governanceEpochDate  compensationAsOfEpochDate  \
```

```
0         1.768176e+09                1.767139e+09
1         1.768176e+09                1.735603e+09
2         1.768176e+09                1.767139e+09

                                                          irWebsite  \
0  http://phx.corporate-ir.net/phoenix.zhtml?c=116466&p=irol-IRHome
1                                                               NaN
2                                       http://investor.apple.com/

  executiveTeam   maxAge  priceHint  previousClose     open    dayLow  \
0            []  86400.0        2.0         178.07  179.050  178.4000
1            []  86400.0        2.0         322.00  320.865  319.3518
2            []  86400.0        2.0         246.70  248.710  245.1900

    dayHigh  regularMarketPreviousClose  regularMarketOpen  \
0  185.3790                      178.07            179.050
1  332.4799                      322.00            320.865
2  251.5600                      246.70            248.710

   regularMarketDayLow  regularMarketDayHigh  dividendRate  dividendYield  \
0             178.4000              185.3790          0.04           0.02
1             319.3518              332.4799          0.84           0.26
2             245.1900              251.5600          1.04           0.42

   exDividendDate  payoutRatio  fiveYearAvgDividendYield   beta  trailingPE  \
0    1.764806e+09       0.0099                      0.06  2.314   45.376240
1    1.765152e+09       0.0809                       NaN  1.086   32.384613
2    1.762733e+09       0.1367                      0.52  1.093   33.241610

   forwardPE       volume  regularMarketVolume  averageVolume  \
0  23.920063  198539922.0          198539922.0    186131977.0
1  29.139809   35014518.0           35014518.0     36498273.0
2  27.102331   54151253.0           54151253.0     46047281.0

   averageVolume10days  averageDailyVolume10Day     bid     ask  bidSize  \
0          170859330.0              170859330.0  182.93  183.39     20.0
1           32441880.0               32441880.0  328.10  328.45      3.0
2           51370470.0               51370470.0  247.43  261.14      4.0

   askSize            marketCap  fiftyTwoWeekLow  fiftyTwoWeekHigh  \
0      1.0   4,463,292,579,840            86.62            212.19
1      1.0   3,977,378,529,280           140.53            340.49
2      1.0   3,659,363,778,560           169.21            288.62

   allTimeHigh  allTimeLow  priceToSalesTrailing12Months  fiftyDayAverage  \
0       212.19    0.033333                     23.849764         184.2150
1       340.49    2.401401                     10.318096         310.1148
```

```
2       288.62    0.049107                      8.793144         271.0410

   twoHundredDayAverage  trailingAnnualDividendRate  \
0             164.39460                        0.04
1             227.18805                        1.02
2             234.16930                        1.02

   trailingAnnualDividendYield currency tradeable  enterpriseValue  \
0                     0.000225      USD     False     4.277320e+12
1                     0.003168      USD     False     3.831270e+12
2                     0.004135      USD     False     3.703010e+12

   profitMargins   floatShares  sharesOutstanding  sharesShort  \
0        0.53007  2.333067e+10       2.430500e+10  256762405.0
1        0.32233  1.079840e+10       5.818000e+09   82670176.0
2        0.26915  1.475035e+10       1.469793e+10  112732788.0

   sharesShortPriorMonth  sharesShortPreviousMonthDate  dateShortInterest  \
0            232898275.0                  1.764288e+09       1.767139e+09
1             79334780.0                  1.764288e+09       1.767139e+09
2            129458559.0                  1.764288e+09       1.767139e+09

   sharesPercentSharesOut  heldPercentInsiders  heldPercentInstitutions  \
0                  0.0106              0.04329                  0.69491
1                  0.0069              0.00568                  0.81179
2                  0.0076              0.01702                  0.64820

   shortRatio  shortPercentOfFloat  impliedSharesOutstanding  bookValue  \
0        1.56               0.0110              2.434700e+10      4.892
1        2.68               0.0142              1.211212e+10     32.033
2        2.68               0.0076              1.477635e+10      4.991

   priceToBook  lastFiscalYearEnd  nextFiscalYearEnd  mostRecentQuarter  \
0    37.473427       1.737850e+09       1.769386e+09       1.761437e+09
1    10.251303       1.735603e+09       1.767139e+09       1.759190e+09
2    49.619312       1.758931e+09       1.790467e+09       1.758931e+09

   earningsQuarterlyGrowth  netIncomeToCommon  trailingEps  forwardEps  \
0                    0.653       9.919800e+10         4.04     7.66386
1                    0.330       1.242510e+11        10.14    11.26912
2                    0.864       1.120100e+11         7.45     9.13759

  lastSplitFactor  lastSplitDate  enterpriseToRevenue  enterpriseToEbitda  \
0           10:01   1.717978e+09               22.856              37.954
1           20:01   1.658102e+09                9.939              26.391
2            4:01   1.598832e+09                8.898              25.582
```

```
   52WeekChange  SandP52WeekChange  lastDividendValue  lastDividendDate  \
0      0.210784           0.116735               0.01      1.764806e+09
1      0.623229           0.116735               0.21      1.765152e+09
2      0.102176           0.116735               0.26      1.762733e+09

  quoteType  currentPrice  targetHighPrice  …  operatingMargins  \
0    EQUITY        183.32            352.0  …           0.63169
1    EQUITY        328.38            432.0  …           0.30512
2    EQUITY        247.65            350.0  …           0.31647

   financialCurrency  symbol.1  language region  typeDisp  \
0                USD      NVDA     en-US     US    Equity
1                USD     GOOGL     en-US     US    Equity
2                USD      AAPL     en-US     US    Equity

          quoteSourceName  triggerable  customPriceAlertConfidence  \
0  Nasdaq Real Time Price         True                        HIGH
1  Nasdaq Real Time Price         True                        HIGH
2  Nasdaq Real Time Price         True                        HIGH

   marketState            longName  sourceInterval  exchangeDataDelayedBy  \
0         POST  NVIDIA Corporation            15.0                    0.0
1         POST       Alphabet Inc.            15.0                    0.0
2         POST          Apple Inc.            15.0                    0.0

           prevName  nameChangeDate  averageAnalystRating  cryptoTradeable  \
0  Usual Stablecoin         1/21/26      1.3 - Strong Buy            False
1               NaN             NaN      1.4 - Strong Buy            False
2               NaN             NaN             2.0 - Buy            False

            shortName  corporateActions  postMarketTime  regularMarketTime  \
0  NVIDIA Corporation                []    1.769039e+09       1.769029e+09
1       Alphabet Inc.                []    1.769039e+09       1.769029e+09
2          Apple Inc.                []    1.769039e+09       1.769029e+09

   exchange  messageBoardId  exchangeTimezoneName  exchangeTimezoneShortName  \
0       NMS     finmb_32307      America/New_York                        EST
1       NMS     finmb_29096      America/New_York                        EST
2       NMS     finmb_24937      America/New_York                        EST

  gmtOffSetMilliseconds     market esgPopulated hasPrePostMarketData  \
0           -18000000.0  us_market        False                 True
1           -18000000.0  us_market        False                 True
2           -18000000.0  us_market        False                 True

  firstTradeDateMilliseconds postMarketChangePercent postMarketPrice  \
0               9.170150e+11                0.139095         183.575
```

```
1               1.092920e+12                0.240578         329.170
2               3.454790e+11                0.327080         248.460

  postMarketChange regularMarketChange regularMarketDayRange  \
0         0.254990            5.250000       178.4 - 185.379
1         0.790009            6.380005   319.3518 - 332.4799
2         0.810013            0.949997       245.19 - 251.56

   fullExchangeName  averageDailyVolume3Month fiftyTwoWeekLowChange  \
0          NasdaqGS               186131977.0             96.700005
1          NasdaqGS                36498273.0            187.850000
2          NasdaqGS                46047281.0             78.439990

  fiftyTwoWeekLowChangePercent fiftyTwoWeekRange fiftyTwoWeekHighChange  \
0                     1.116370    86.62 - 212.19             -28.869995
1                     1.336725   140.53 - 340.49             -12.109985
2                     0.463566   169.21 - 288.62             -40.970000

  fiftyTwoWeekHighChangePercent fiftyTwoWeekChangePercent  dividendDate  \
0                     -0.136057                 21.078396  1.766707e+09
1                     -0.035566                 62.322926  1.765757e+09
2                     -0.141951                 10.217571  1.762992e+09

   earningsTimestamp earningsTimestampStart earningsTimestampEnd  \
0       1.772053e+09           1.772053e+09         1.772053e+09
1       1.770239e+09           1.770239e+09         1.770239e+09
2       1.769720e+09           1.769720e+09         1.769720e+09

  earningsCallTimestampStart earningsCallTimestampEnd  isEarningsDateEstimate  \
0               1.763590e+09             1.763590e+09                   False
1               1.770241e+09             1.770241e+09                   False
2               1.769724e+09             1.769724e+09                   False

  epsTrailingTwelveMonths epsForward epsCurrentYear  priceEpsCurrentYear  \
0                    4.04    7.66386        4.69210            39.069927
1                   10.14   11.26912       10.54084            31.153114
2                    7.45    9.13759        8.24255            30.045313

   fiftyDayAverageChange  fiftyDayAverageChangePercent  \
0              -0.894989                     -0.004858
1              18.265198                      0.058898
2             -23.390991                     -0.086301

   twoHundredDayAverageChange  twoHundredDayAverageChangePercent  \
0                   18.925400                           0.115122
1                  101.191956                           0.445411
2                   13.480698                           0.057568
```

```
  regularMarketChangePercent regularMarketPrice  displayName  \
0                   2.948279             183.32       NVIDIA
1                   1.981368             328.38          NaN
2                   0.385082             247.65        Apple

   trailingPegRatio  ipoExpectedDate yield  totalAssets  navPrice  category  \
0            0.6722              NaN   NaN          NaN       NaN       NaN
1            1.7001              NaN   NaN          NaN       NaN       NaN
2            2.4965              NaN   NaN          NaN       NaN       NaN

   ytdReturn  beta3Year  fundFamily  fundInceptionDate  legalType  \
0        NaN        NaN         NaN                NaN        NaN
1        NaN        NaN         NaN                NaN        NaN
2        NaN        NaN         NaN                NaN        NaN

   threeYearAverageReturn fiveYearAverageReturn  trailingThreeMonthReturns  \
0                     NaN                   NaN                        NaN
1                     NaN                   NaN                        NaN
2                     NaN                   NaN                        NaN

   trailingThreeMonthNavReturns  netAssets  netExpenseRatio  address2  fax  \
0                           NaN        NaN              NaN       NaN  NaN
1                           NaN        NaN              NaN       NaN  NaN
2                           NaN        NaN              NaN       NaN  NaN

   industrySymbol  underlyingSymbol  firstTradeDateEpochUtc  timeZoneFullName  \
0             NaN               NaN                     NaN               NaN
1             NaN               NaN                     NaN               NaN
2             NaN               NaN                     NaN               NaN

  timeZoneShortName  uuid newSymbol  openInterest  prevExchange  \
0               NaN   NaN       NaN           NaN           NaN
1               NaN   NaN       NaN           NaN           NaN
2               NaN   NaN       NaN           NaN           NaN

   exchangeTransferDate address3  morningStarOverallRating  \
0                   NaN      NaN                       NaN
1                   NaN      NaN                       NaN
2                   NaN      NaN                       NaN

   morningStarRiskRating annualReportExpenseRatio  lastCapGain  \
0                    NaN                      NaN          NaN
1                    NaN                      NaN          NaN
2                    NaN                      NaN          NaN

  annualHoldingsTurnover  prevTicker  tickerChangeDate  newListingDate  \
```

```
0                    NaN         NaN                NaN             NaN
1                    NaN         NaN                NaN             NaN
2                    NaN         NaN                NaN             NaN

   delistingDate
0            NaN
1            NaN
2            NaN

[3 rows x 224 columns]
```

Detect key NLP columns robustly

[23]:
```
def find_col(candidates, columns):
    colmap = {c.lower(): c for c in columns}
    for cand in candidates:
        if cand.lower() in colmap:
            return colmap[cand.lower()]
    return None

sector_col = find_col(["sector", "sectorDisp", "sectorKey"], df.columns)
industry_col = find_col(["industry", "industryDisp", "industryKey"], df.columns)
summary_col = find_col(["longBusinessSummary", "businessSummary", "summary",␣
 ↪"description"], df.columns)

symbol_col = find_col(["symbol", "ticker"], df.columns)
name_col = find_col(["company_name", "shortName", "longName", "name"], df.
 ↪columns)
quote_type_col = find_col(["quoteType"], df.columns)

print("Detected columns:")
print("  symbol:", symbol_col)
print("  company:", name_col)
print("  quoteType:", quote_type_col)
print("  sector:", sector_col)
print("  industry:", industry_col)
print("  summary:", summary_col)

assert sector_col and industry_col and summary_col, "Missing required NLP␣
 ↪columns."
```

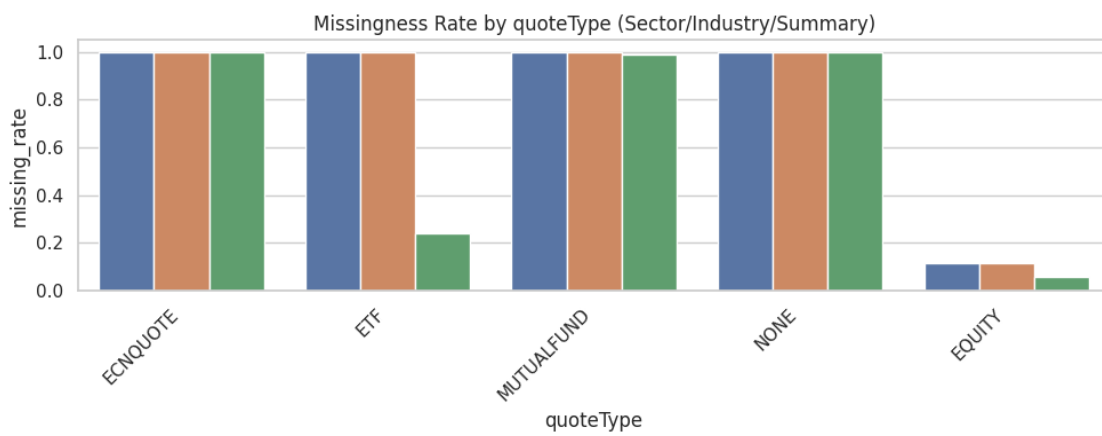
```
Detected columns:
  symbol: symbol
  company: company_name
  quoteType: quoteType
  sector: sector
  industry: industry
  summary: longBusinessSummary
```

Coverage / missingness (overall + by quoteType)

[24]:
```
def missing_report(s: pd.Series, name: str):
    s2 = s.copy()
    null_count = int(s2.isna().sum())
    empty_count = int((s2.fillna("").astype(str).str.strip() == "").sum())
    n = len(s2)
    return {
        "field": name,
        "rows": n,
        "null_count": null_count,
        "empty_or_whitespace_count": empty_count,
        "nonempty_count": n - empty_count,
        "pct_null": null_count / n,
        "pct_empty_or_whitespace": empty_count / n,
    }

coverage = pd.DataFrame([
    missing_report(df[sector_col], sector_col),
    missing_report(df[industry_col], industry_col),
    missing_report(df[summary_col], summary_col),
])

display(coverage)


if quote_type_col:
    def is_missing_text(series):
        return series.isna() | (series.astype(str).str.strip() == "")

    tmp = df.copy()
    tmp["_miss_sector"] = is_missing_text(tmp[sector_col])
    tmp["_miss_industry"] = is_missing_text(tmp[industry_col])
    tmp["_miss_summary"] = is_missing_text(tmp[summary_col])

    by_qt = (tmp.
 ↪groupby(quote_type_col)[["_miss_sector","_miss_industry","_miss_summary"]]
                .mean()
                .sort_values("_miss_sector", ascending=False)
                .reset_index())

    display(by_qt)


    plt.figure(figsize=(10,4))
    melted = by_qt.melt(id_vars=quote_type_col, var_name="field",␣
 ↪value_name="missing_rate")
```

```
    sns.barplot(data=melted, x=quote_type_col, y="missing_rate", hue="field")
    plt.xticks(rotation=45, ha="right")
    plt.title("Missingness Rate by quoteType (Sector/Industry/Summary)")
    plt.tight_layout()
    plt.show()
```

```
                 field   rows  null_count  empty_or_whitespace_count  \
0               sector  10284        1812                       1812
1             industry  10284        1812                       1812
2  longBusinessSummary  10284        1103                       1103

   nonempty_count  pct_null  pct_empty_or_whitespace
0            8472  0.176196                 0.176196
1            8472  0.176196                 0.176196
2            9181  0.107254                 0.107254

    quoteType  _miss_sector  _miss_industry  _miss_summary
0    ECNQUOTE      1.000000        1.000000       1.000000
1         ETF      1.000000        1.000000       0.241206
2  MUTUALFUND      1.000000        1.000000       0.990566
3        NONE      1.000000        1.000000       1.000000
4      EQUITY      0.114178        0.114178       0.055939
```



Sector/industry distributions + cross-checks

[25]:
```
sector_counts = df[sector_col].value_counts(dropna=False).reset_index()
sector_counts.columns = ["sector", "count"]
industry_counts = df[industry_col].value_counts(dropna=False).reset_index()
industry_counts.columns = ["industry", "count"]

print("Unique sectors:", df[sector_col].nunique(dropna=True))
print("Unique industries:", df[industry_col].nunique(dropna=True))
```

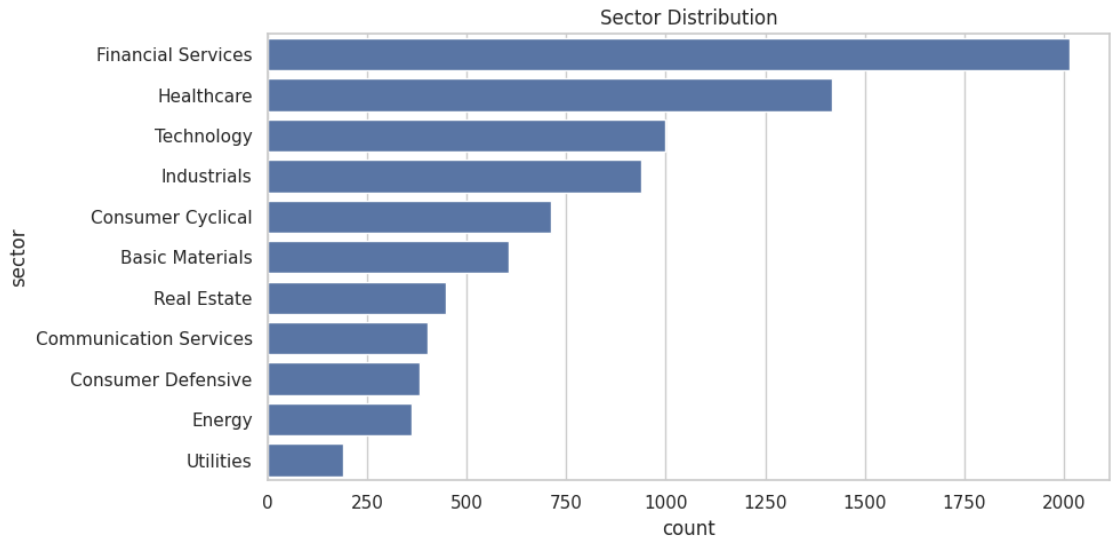
```python
display(sector_counts.head(20))
display(industry_counts.head(20))

# Plot sector distribution
plt.figure(figsize=(10,5))
top_sector = sector_counts.dropna().head(20)
sns.barplot(data=top_sector, x="count", y="sector")
plt.title("Sector Distribution")
plt.tight_layout()
plt.show()

# Industries appearing in multiple sectors (potential label issues)
inds_multi_sector = (df.dropna(subset=[sector_col, industry_col])
                       .groupby(industry_col)[sector_col]
                       .nunique()
                       .sort_values(ascending=False)
                       .reset_index()
                       .rename(columns={sector_col:"n_sectors_for_industry"}))

display(inds_multi_sector.head(30))
```

```
Unique sectors: 11
Unique industries: 145
```

	sector	count
0	Financial Services	2014
1	NaN	1812
2	Healthcare	1418
3	Technology	1000
4	Industrials	939
5	Consumer Cyclical	711
6	Basic Materials	607
7	Real Estate	448
8	Communication Services	401
9	Consumer Defensive	381
10	Energy	362
11	Utilities	191

	industry	count
0	NaN	1812
1	Biotechnology	721
2	Asset Management	569
3	Banks - Regional	484
4	Shell Companies	450
5	Software - Application	320
6	Software - Infrastructure	197
7	Medical Devices	182
8	Other Industrial Metals & Mining	175
9	Drug Manufacturers - Specialty & Generic	163

10	Gold	128
11	Capital Markets	121
12	Oil & Gas E&P	120
13	Packaged Foods	108
14	Aerospace & Defense	104
15	Telecom Services	104
16	Specialty Industrial Machinery	102
17	Internet Content & Information	99
18	REIT - Mortgage	99
19	Oil & Gas Midstream	93



	industry	n_sectors_for_industry
0	Advertising Agencies	1
1	Aerospace & Defense	1
2	Agricultural Inputs	1
3	Airlines	1
4	Airports & Air Services	1
5	Aluminum	1
6	Apparel Manufacturing	1
7	Apparel Retail	1
8	Asset Management	1
9	Auto & Truck Dealerships	1
10	Auto Manufacturers	1
11	Auto Parts	1
12	Banks - Diversified	1
13	Banks - Regional	1
14	Beverages - Brewers	1
15	Beverages - Non-Alcoholic	1
16	Beverages - Wineries & Distilleries	1

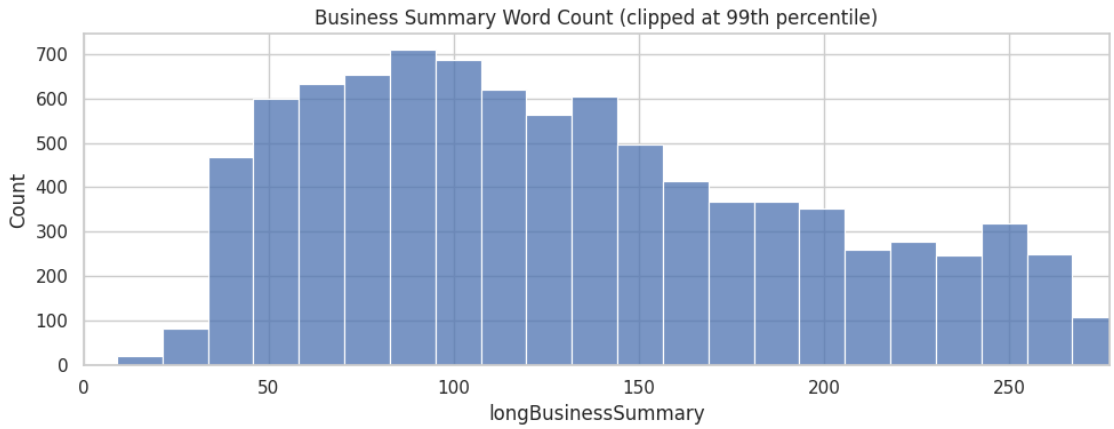
```
17                        Biotechnology                        1
18                         Broadcasting                        1
19                    Building Materials                       1
20         Building Products & Equipment                       1
21         Business Equipment & Supplies                       1
22                      Capital Markets                        1
23                            Chemicals                        1
24                          Coking Coal                        1
25               Communication Equipment                       1
26                     Computer Hardware                       1
27                         Confectioners                       1
28                         Conglomerates                       1
29                   Consulting Services                       1
```

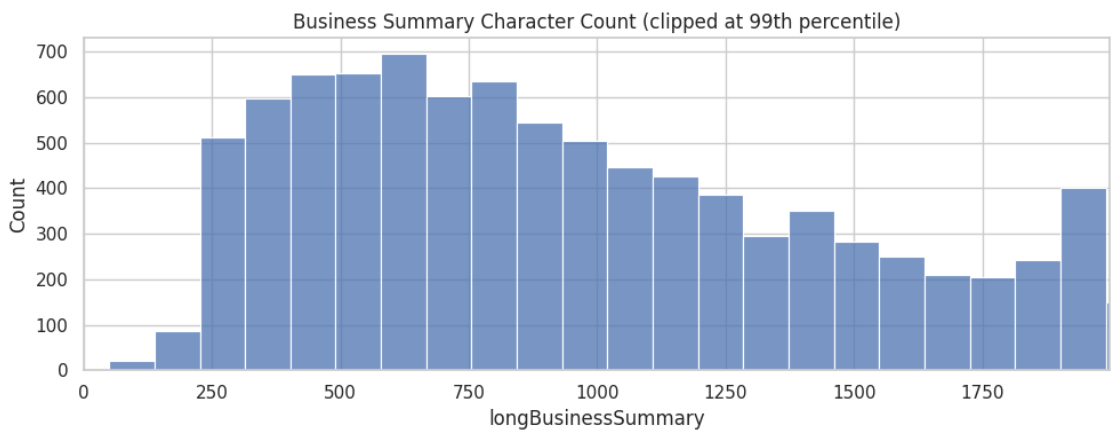
Business summary length profiling (plots + outliers)

[26]:
```
summ = df[summary_col].fillna("").astype(str).str.strip()
nonempty = summ[summ != ""]

char_len = nonempty.str.len()
word_len = nonempty.str.split().str.len()
sent_len = nonempty.str.count(r"[.!?]") + 1

length_stats = pd.DataFrame({
    "metric": ["char_len","word_len","sent_len"],
    "count": [len(char_len), len(word_len), len(sent_len)],
    "mean": [char_len.mean(), word_len.mean(), sent_len.mean()],
    "median": [char_len.median(), word_len.median(), sent_len.median()],
    "p95": [char_len.quantile(0.95), word_len.quantile(0.95), sent_len.
 ↪quantile(0.95)],
    "p99": [char_len.quantile(0.99), word_len.quantile(0.99), sent_len.
 ↪quantile(0.99)],
    "max": [char_len.max(), word_len.max(), sent_len.max()],
})

display(length_stats)

# Histograms
plt.figure(figsize=(10,4))
sns.histplot(word_len, bins=60)
plt.xlim(0, np.percentile(word_len, 99))
plt.title("Business Summary Word Count (clipped at 99th percentile)")
plt.tight_layout()
plt.show()

plt.figure(figsize=(10,4))
sns.histplot(char_len, bins=60)
plt.xlim(0, np.percentile(char_len, 99))
```

```
plt.title("Business Summary Character Count (clipped at 99th percentile)")
plt.tight_layout()
plt.show()

# Very short summaries for inspection
very_short_idx = nonempty.index[word_len < 15]
short_examples = df.loc[very_short_idx, [c for c in [symbol_col, name_col,␣
 ↪quote_type_col, sector_col, industry_col, summary_col] if c]].head(20)
display(short_examples)
```

```
     metric  count        mean  median     p95     p99   max
0  char_len   9181  966.489271   864.0  1944.0  1998.0  5342
1  word_len   9181  133.112624   122.0   254.0   277.0   746
2  sent_len   9181    8.585775     8.0    14.0    18.0    38
```






```
     symbol                                          company_name quoteType  \
3022   MEGI  NYLI CBRE Global Infrastructure Megatrends Term Fund    EQUITY
```

```
3669   XFLT  XAI Octagon Floating Rate & Alternative Income Trust    EQUITY
4544   OCCI                              OFS Credit Company, Inc.    EQUITY
9043  BPYPM                     Brookfield Property Partners L.P.    EQUITY
9230  OCCIM                              OFS Credit Company, Inc.    EQUITY
9231  OCCIN                              OFS Credit Company, Inc.    EQUITY
9232  OCCIO                              OFS Credit Company, Inc.    EQUITY

                  sector          industry  \
3022  Financial Services  Asset Management
3669  Financial Services  Asset Management
4544  Financial Services  Asset Management
9043  Financial Services  Asset Management
9230  Financial Services  Asset Management
9231  Financial Services  Asset Management
9232  Financial Services  Asset Management


                                                                        ␣
 ↪longBusinessSummary
3022  NYLI CBRE Global Infrastructure Megatrends Term Fund is headquartered in␣
 ↪New York, New York.
3669   XAI Octagon Floating Rate & Alternative Income Trust is headquartered in␣
 ↪Chicago, Illinois.
4544                                          OFS Credit Company, Inc. is a␣
 ↪fund of OFS Advisor.
9043                     Brookfield Property Preferred L.P. is headquartered in␣
 ↪Hamilton, Bermuda.
9230                                          OFS Credit Company, Inc. is a␣
 ↪fund of OFS Advisor.
9231                                          OFS Credit Company, Inc. is a␣
 ↪fund of OFS Advisor.
9232                                          OFS Credit Company, Inc. is a␣
 ↪fund of OFS Advisor.
```

— Boilerplate phrase checks + top terms/ngrams

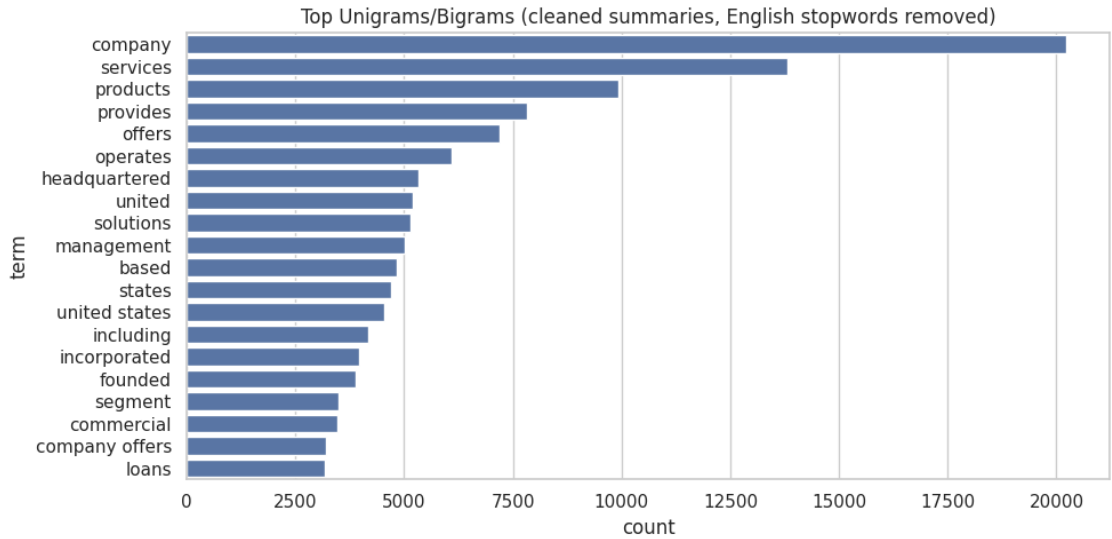
[27]:
```python
def basic_clean(text: str) -> str:
    text = str(text).lower()
    text = re.sub(r"[^a-z0-9\s]", " ", text)
    text = re.sub(r"\s+", " ", text).strip()
    return text

clean_texts = nonempty.map(basic_clean)
summ_lower = nonempty.str.lower()

common_phrases = [
    "headquartered in",
    "was founded in",
```

```
    "incorporated in",
    "together with its subsidiaries",
    "engages in",
    "operates in",
    "provides",
]

phrase_freq = pd.DataFrame({
    "phrase": common_phrases,
    "count": [int(summ_lower.str.contains(p).sum()) for p in common_phrases],
})
phrase_freq["pct_of_nonempty"] = phrase_freq["count"] / len(nonempty)
display(phrase_freq.sort_values("count", ascending=False))

# Top unigrams/bigrams
vec = CountVectorizer(stop_words="english", max_features=60, ngram_range=(1,2))
X = vec.fit_transform(clean_texts)
terms = np.array(vec.get_feature_names_out())
freq = np.asarray(X.sum(axis=0)).ravel()

top_idx = freq.argsort()[::-1][:30]
top_terms = pd.DataFrame({"term": terms[top_idx], "count": freq[top_idx]})
display(top_terms)

plt.figure(figsize=(10,5))
sns.barplot(data=top_terms.head(20), x="count", y="term")
plt.title("Top Unigrams/Bigrams (cleaned summaries, English stopwords removed)")
plt.tight_layout()
plt.show()
```

```
                          phrase  count  pct_of_nonempty
0                headquartered in   5341         0.581745
6                        provides   4520         0.492321
1                  was founded in   3866         0.421087
2                 incorporated in   3768         0.410413
4                      engages in   2229         0.242784
3  together with its subsidiaries   1196         0.130269
5                     operates in    806         0.087790
```

```
              term  count
0          company  20212
1         services  13808
2         products   9931
3         provides   7837
4           offers   7202
5         operates   6089
6    headquartered   5342
7           united   5208
```

```
8        solutions   5138
9       management   5011
10           based   4832
11          states   4699
12   united states   4543
13       including   4179
14    incorporated   3969
15         founded   3883
16         segment   3480
17      commercial   3453
18  company offers   3201
19           loans   3188
20        business   3163
21         systems   3005
22           known   2736
23     development   2694
24      technology   2641
25      investment   2584
26     corporation   2490
27         engages   2458
28            fund   2454
29        addition   2442
```



Duplicate summaries (exact) + near-duplicates (TF-IDF)

[28]:
```
# Exact duplicates
summary_stripped = df[summary_col].fillna("").astype(str).str.strip()
nonempty_mask = summary_stripped != ""
```

```
dup_counts = summary_stripped[nonempty_mask].value_counts()
dup_summary_stats = pd.DataFrame([{
    "distinct_nonempty_summaries": int(dup_counts.shape[0]),
    "distinct_duplicated_summaries": int((dup_counts > 1).sum()),
    "rows_in_duplicated_summaries": int(summary_stripped[nonempty_mask].
 ↪map(dup_counts).gt(1).sum()),
    "max_repetition_single_summary": int(dup_counts.max())
}])

display(dup_summary_stats)

top_dup = dup_counts.head(10).reset_index()
top_dup.columns = ["summary_text", "count"]
top_dup["preview"] = top_dup["summary_text"].str.slice(0, 160) + "..."
display(top_dup[["count","preview"]])

# Near-duplicates with TF-IDF + NearestNeighbors
SAMPLE_N = None
rng = np.random.default_rng(42)

idx = summary_stripped[nonempty_mask].index.to_numpy()
if SAMPLE_N is not None and SAMPLE_N < len(idx):
    idx = rng.choice(idx, size=SAMPLE_N, replace=False)

texts = df.loc[idx, summary_col].fillna("").astype(str).map(basic_clean)

tfidf = TfidfVectorizer(stop_words="english", min_df=3, max_df=0.90,␣
 ↪ngram_range=(1,2))
X = tfidf.fit_transform(texts)

nn = NearestNeighbors(n_neighbors=6, metric="cosine")
nn.fit(X)
dist, neigh = nn.kneighbors(X)

SIM_THRESHOLD = 0.90
pairs = []
for row_i in range(neigh.shape[0]):
    i = idx[row_i]
    for k in range(1, neigh.shape[1]):
        j = idx[neigh[row_i, k]]
        sim = 1 - dist[row_i, k]
        if sim >= SIM_THRESHOLD:
            pairs.append((i, j, sim))

near_dup_df = (pd.DataFrame(pairs, columns=["row_i","row_j","tfidf_cosine_sim"])
               .sort_values("tfidf_cosine_sim", ascending=False))
```

```
display(near_dup_df.head(30))

# add metadata for top pairs
cols_to_show = [c for c in [symbol_col, name_col, quote_type_col, sector_col,␣
↪industry_col, summary_col] if c]
if len(near_dup_df) > 0 and cols_to_show:
    sample_pairs = near_dup_df.head(20).copy()
    left = df.loc[sample_pairs["row_i"], cols_to_show].reset_index().
↪rename(columns={"index":"row_i"})
    right = df.loc[sample_pairs["row_j"], cols_to_show].reset_index().
↪rename(columns={"index":"row_j"})
    show = sample_pairs.merge(left, on="row_i").merge(right, on="row_j",␣
↪suffixes=("_A","_B"))
    display(show.head(10))
```

```
   distinct_nonempty_summaries  distinct_duplicated_summaries  \
0                         7628                           1044

   rows_in_duplicated_summaries  max_repetition_single_summary
0                          2597                             22

   count  \
0     22
1     16
2     15
3     15
4     13
5     11
6     10
7      8
8      7
9      7


                                                                                 ␣
↪                                                                                ␣
↪preview
0  Federal Home Loan Mortgage Corporation operates in the secondary mortgage␣
↪market in the United States. It operates through two segments, Single-Family␣
↪and Multi…
1  Federal National Mortgage Association provides financing solutions for␣
↪residential mortgages in the United States. The company operates through␣
↪Single-Family an…
2  Public Storage, a member of the S&P 500, is a REIT that primarily acquires,␣
↪develops, owns, and operates self-storage facilities. At September 30, 2025,␣
↪we: (i)…
3  Bank of America Corporation, through its subsidiaries, provides various␣
↪financial products and services for individual consumers, small and␣
↪middle-market busine…
```

```
4  The Connecticut Light and Power Company, a regulated electric utility,␣
↪engages in the purchase, delivery, and sale of electricity to residential,␣
↪commercial, an…
5  Enbridge Inc., together with its subsidiaries, operates as an energy␣
↪infrastructure company. The company operates through four segments: Liquids␣
↪Pipelines, Gas …
6  Morgan Stanley, a financial holding company, provides various financial␣
↪products and services to governments, financial institutions, and individuals␣
↪in the Ame…
7  Union Electric Company, doing business as Ameren Missouri, engages in the␣
↪rate-regulated electric generation, transmission, and distribution business in␣
↪Missour…
8  Adamas Trust, Inc. acquires, invests in, finances, and manages␣
↪mortgage-related single-family and multi-family residential assets in the␣
↪United States. The comp…
9  Federal Agricultural Mortgage Corporation provides a secondary market for␣
↪various loans made to borrowers in the United States. It operates through␣
↪seven segmen…
```

	row_i	row_j	tfidf_cosine_sim
4745	10283	7494	1.0
0	1	7497	1.0
4743	10276	10276	1.0
2	11	7504	1.0
3	11	7505	1.0
4	11	8365	1.0
5	11	8362	1.0
6	11	8363	1.0
4724	10219	3841	1.0
4723	10215	10215	1.0
4722	10214	10214	1.0
4721	10214	4157	1.0
4720	10213	10214	1.0
4719	10213	4157	1.0
39	77	7527	1.0
38	73	73	1.0
37	73	7579	1.0
35	66	7522	1.0
34	54	8627	1.0
33	48	7511	1.0
31	45	7523	1.0
28	40	7509	1.0
26	36	10132	1.0
25	35	7506	1.0
4706	10204	10203	1.0
4705	10204	10204	1.0
4704	10203	10203	1.0
4703	10203	10204	1.0

```
4702  10201  10201               1.0
4701  10198  10198               1.0

   row_i  row_j  tfidf_cosine_sim symbol_A        company_name_A quoteType_A  \
0  10283   7494               1.0    MTPLF       Metaplanet Inc.      EQUITY
1      1   7497               1.0    GOOGL         Alphabet Inc.      EQUITY
2  10276  10276               1.0    MAWAF  Maruwa Co., Ltd./ADR      EQUITY
3     11   7504               1.0      JPM   JPMORGAN CHASE & CO      EQUITY
4     11   7504               1.0      JPM   JPMORGAN CHASE & CO      EQUITY
5     11   7504               1.0      JPM   JPMORGAN CHASE & CO      EQUITY
6     11   7504               1.0      JPM   JPMORGAN CHASE & CO      EQUITY
7     11   7504               1.0      JPM   JPMORGAN CHASE & CO      EQUITY
8     11   7505               1.0      JPM   JPMORGAN CHASE & CO      EQUITY
9     11   7505               1.0      JPM   JPMORGAN CHASE & CO      EQUITY

                 sector_A                     industry_A  \
0       Consumer Cyclical                    Restaurants
1  Communication Services  Internet Content & Information
2             Technology           Electronic Components
3      Financial Services             Banks - Diversified
4      Financial Services             Banks - Diversified
5      Financial Services             Banks - Diversified
6      Financial Services             Banks - Diversified
7      Financial Services             Banks - Diversified
8      Financial Services             Banks - Diversified
9      Financial Services             Banks - Diversified


 longBusinessSummary_A  \
0  Metaplanet Inc. engages in hotel management operation and development in Japan. The company operates through Hotel Business and Bitcoin Treasury Business segments. It is also i…
1  Alphabet Inc. offers various products and platforms in the United States, Europe, the Middle East, Africa, the Asia-Pacific, Canada, and Latin America. It operates through Goog…
2  Maruwa Co.,Ltd. produces and sells ceramics and electronic parts in Japan and internationally. The Company has two reportable segments: Lamic Components Business, and Lighting …
3  JPMorgan Chase & Co. operates as a financial services company worldwide. It operates through three segments: Consumer & Community Banking, Commercial & Investment Bank, and Ass…
4  JPMorgan Chase & Co. operates as a financial services company worldwide. It operates through three segments: Consumer & Community Banking, Commercial & Investment Bank, and Ass…
5  JPMorgan Chase & Co. operates as a financial services company worldwide. It operates through three segments: Consumer & Community Banking, Commercial & Investment Bank, and Ass…
```

```
6  JPMorgan Chase & Co. operates as a financial services company worldwide. It␣
↪operates through three segments: Consumer & Community Banking, Commercial &␣
↪Investment Bank, and Ass…
7  JPMorgan Chase & Co. operates as a financial services company worldwide. It␣
↪operates through three segments: Consumer & Community Banking, Commercial &␣
↪Investment Bank, and Ass…
8  JPMorgan Chase & Co. operates as a financial services company worldwide. It␣
↪operates through three segments: Consumer & Community Banking, Commercial &␣
↪Investment Bank, and Ass…
9  JPMorgan Chase & Co. operates as a financial services company worldwide. It␣
↪operates through three segments: Consumer & Community Banking, Commercial &␣
↪Investment Bank, and Ass…

  symbol_B        company_name_B quoteType_B                sector_B  \
0    MPJPY       Metaplanet Inc.      EQUITY                     NaN
1     GOOG         Alphabet Inc.      EQUITY  Communication Services
2    MAWAF  Maruwa Co., Ltd./ADR      EQUITY              Technology
3   JPM-PC   JPMORGAN CHASE & CO      EQUITY      Financial Services
4   JPM-PC   JPMORGAN CHASE & CO      EQUITY      Financial Services
5   JPM-PC   JPMORGAN CHASE & CO      EQUITY      Financial Services
6   JPM-PC   JPMORGAN CHASE & CO      EQUITY      Financial Services
7   JPM-PC   JPMORGAN CHASE & CO      EQUITY      Financial Services
8   JPM-PD   JPMORGAN CHASE & CO      EQUITY      Financial Services
9   JPM-PD   JPMORGAN CHASE & CO      EQUITY      Financial Services

                      industry_B  \
0                            NaN
1  Internet Content & Information
2           Electronic Components
3             Banks - Diversified
4             Banks - Diversified
5             Banks - Diversified
6             Banks - Diversified
7             Banks - Diversified
8             Banks - Diversified
9             Banks - Diversified

                                                                          ␣
↪                                                                         ␣
↪ longBusinessSummary_B
0  Metaplanet Inc. engages in hotel management operation and development in␣
↪Japan. The company operates through Hotel Business and Bitcoin Treasury␣
↪Business segments. It is also i…
1  Alphabet Inc. offers various products and platforms in the United States,␣
↪Europe, the Middle East, Africa, the Asia-Pacific, Canada, and Latin America.␣
↪It operates through Goog…
```

```
2  Maruwa Co.,Ltd. produces and sells ceramics and electronic parts in Japan and␣
↪internationally. The Company has two reportable segments: Lamic Components␣
↪Business, and Lighting …
3  JPMorgan Chase & Co. operates as a financial services company worldwide. It␣
↪operates through three segments: Consumer & Community Banking, Commercial &␣
↪Investment Bank, and Ass…
4  JPMorgan Chase & Co. operates as a financial services company worldwide. It␣
↪operates through three segments: Consumer & Community Banking, Commercial &␣
↪Investment Bank, and Ass…
5  JPMorgan Chase & Co. operates as a financial services company worldwide. It␣
↪operates through three segments: Consumer & Community Banking, Commercial &␣
↪Investment Bank, and Ass…
6  JPMorgan Chase & Co. operates as a financial services company worldwide. It␣
↪operates through three segments: Consumer & Community Banking, Commercial &␣
↪Investment Bank, and Ass…
7  JPMorgan Chase & Co. operates as a financial services company worldwide. It␣
↪operates through three segments: Consumer & Community Banking, Commercial &␣
↪Investment Bank, and Ass…
8  JPMorgan Chase & Co. operates as a financial services company worldwide. It␣
↪operates through three segments: Consumer & Community Banking, Commercial &␣
↪Investment Bank, and Ass…
9  JPMorgan Chase & Co. operates as a financial services company worldwide. It␣
↪operates through three segments: Consumer & Community Banking, Commercial &␣
↪Investment Bank, and Ass…
```

[29]:

```python
def safe_str(x):
    if pd.isna(x):
        return ""
    return str(x).strip()

def build_embed_text(row):
    parts = []
    if safe_str(row[sector_col]):
        parts.append(f"Sector: {safe_str(row[sector_col])}")
    if safe_str(row[industry_col]):
        parts.append(f"Industry: {safe_str(row[industry_col])}")
    if safe_str(row[summary_col]):
        parts.append(f"Summary: {safe_str(row[summary_col])}")
    return " | ".join(parts).strip()

df["embed_text"] = df.apply(build_embed_text, axis=1)
df["embed_text_is_empty"] = df["embed_text"].str.strip().eq("")

embed_ready_stats = pd.DataFrame([{
    "rows_total": len(df),
    "rows_embed_text_empty": int(df["embed_text_is_empty"].sum()),
    "rows_embed_text_nonempty": int((~df["embed_text_is_empty"]).sum())
```

```
}])

display(embed_ready_stats)
```

```
   rows_total  rows_embed_text_empty  rows_embed_text_nonempty
0       10284                   1071                      9213
```

[30]:
```
# 4) Check what changed
!git status -sb

# 5) Stage + commit
#!git add -A
#!git commit -m "push jupyter hub changes"

# 6) Push
#!git push
```

```
## main…origin/main
```